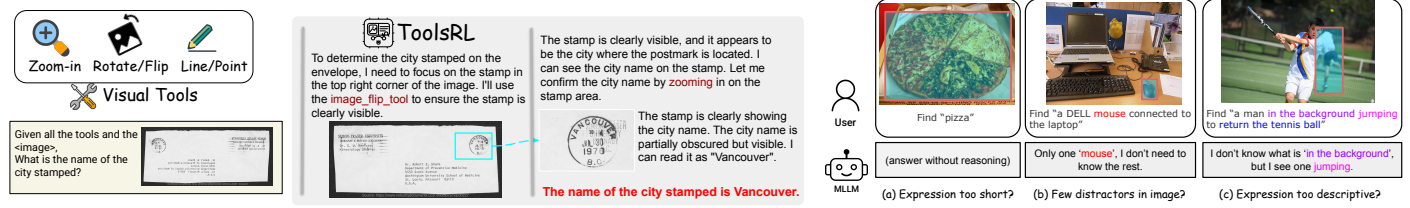


My research centers on **reasoning and understanding with multimodal large language models (MLLMs)**, with the goal of building multimodal AI agents that can operate reliably in real-world environments. Recent LLMs have demonstrated expert-level text-based reasoning in math and coding, and can sustain hours of “thinking” [1, 2, 3]. However, today’s MLLMs still struggle with spatial and fine-grained visual reasoning, capabilities that come naturally to humans [4].

This gap limits real-world deployment. To address this, my work focuses on two directions: (1) **an emerging reasoning paradigm, “Thinking with Images”** and (2) **fine-grained visual reasoning**. Looking ahead, I am also interested in learning from video and streaming inputs, and in mitigating “reasoning collapse” in RLVR, both important for building next-generation MLLMs.



(a) MLLM that can “think with images” with multiple tools, from my intern work in Amazon [5]. (b) Deficits of RefCOCO+/g leading MLLMs to exploit shortcuts, shown in my work [4].

“Thinking with Images”: An Emerging Paradigm in MLLM Reasoning

One critical difference between LLM and MLLM reasoning is that, for LLMs, the reasoning space is the same as the input space (text tokens). In contrast, most MLLMs are trained to *reason* in text subspace while *perceiving* multimodal inputs. This mismatch can make them struggle on tasks that seem simple to humans.

OpenAI o3 [6] is an early example of explicitly reasoning in multimodal space (so-called “Thinking with Images”) by interleaving text with zoomed-in crops during reasoning. This is a meaningful step forward, but the available operations are still limited compared to human visual manipulation (e.g., drawing, rotating). My intern work with Amazon, ToolsRL [5], broadened the visual toolset available to MLLMs, including zooming, drawing, rotating, and flipping, by utilizing synthetic data and curriculum RL training. To my knowledge, it is the first open-source MLLM that supports both multi-tool use and long, multi-round reasoning (3.4 tool calls on average per example). The model achieves SOTA performance across a variety of benchmarks.

Fine-grained Visual Reasoning

MLLMs usually adopt CLIP-like vision encoders. While strong at semantic understanding, they are not optimized for fine-grained localization, e.g., the exact pixel location for objects. One direct approach is to train MLLMs on fine-grained tasks such as detection and segmentation. My intern work at Adobe [7] improved MLLM performance in a fine-grained task: referring expression segmentation. I proposed a new dataset that includes intermediate masks related to the query, enabling step-by-step reasoning with grounding.

Another important observation following this work is that, while RefCOCO+/g [8, 9, 10] are commonly used, they require little reasoning (Fig 1b), and current MLLMs have largely saturated performance on them (90%+ accuracy). In practice, real referring tasks can be much more complex. In my work [4], I analyzed several factors that induce “reasoning shortcuts”: (1) overly simple expressions, (2) a lack of hard distractors, and (3) redundant descriptions. To bridge this gap, I built a new benchmark with challenging distractors and demonstrated that even SOTA MLLMs struggle with complex, fine-grained visual reasoning (< 60% accuracy).

Future Directions: Learning on Video, “Reasoning Collapse” in RLVR

Humans learn richly from video, yet most foundation MLLMs are trained predominantly on static images and text. Compared to images, video provides a natural notion of “next state” (the next frame), which may align well with autoregressive learning. However, video also contains substantial redundancy, motivating new solutions for compressing and representing spatiotemporal information. I am excited to explore effective learning from video and streaming inputs.

Another direction I am enthusiastic about is mitigating “reasoning collapse” in RLVR (Reinforcement Learning with Verifiable Reward) algorithms. The *reasoning collapse* [11] revealed that current RLVR mainly collapse the model distribution onto a narrow solution path that fits the reward, while failing to elicit genuinely new behaviors beyond the base model. I aim to develop RL methods that address this issue, enabling models to transcend the limits of pre-training and improve in new environments.

References

- [1] OpenAI. “GPT-4 Technical Report”. In: *arXiv preprint arXiv:2303.08774* (2023).
- [2] Gemini Team et al. “Gemini: A Family of Highly Capable Multimodal Models”. In: *arXiv preprint arXiv:2312.11805* (2023).
- [3] Zhihong Shao et al. “DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models”. In: *arXiv preprint arXiv:2402.03300* (2024).
- [4] Qihua Dong et al. “Ref-Adv: Exploring MLLM Visual Reasoning in Referring Expression Tasks”. In: *International Conference on Learning Representations (ICLR)*. Under review. 2025. URL: <https://openreview.net/forum?id=iEBgrepr9i>.
- [5] Qihua Dong et al. “Visual Reasoning through Tool-supervised Reinforcement Learning”. In: *Submitted to Proc. CVPR* (2025).
- [6] OpenAI. *OpenAI o3 (Model Documentation)*. 2025. URL: <https://platform.openai.com/docs/models/o3> (visited on 12/22/2025).
- [7] Qihua Dong et al. “CoT Referring: Chain-of-Thought Grounding Improve Localization in Referring Expression Tasks”. In: *Submitted to ICLR* (2025).
- [8] Sahar Kazemzadeh et al. “Referitgame: Referring to objects in photographs of natural scenes”. In: *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*. 2014, pp. 787–798.
- [9] Licheng Yu et al. “Modeling context in referring expressions”. In: *Proceedings of the European Conference on Computer Vision (ECCV)*. Springer. 2016, pp. 69–85.
- [10] Junhua Mao et al. “Generation and comprehension of unambiguous object descriptions”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*. 2016, pp. 11–20.
- [11] Yang Yue et al. “Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model?”. In: *arXiv preprint arXiv:2504.13837* (2025).